Ensemble Entropy Metric for Hyperspectral Anomaly Detection

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***Abstract*— In hyperspectral anomaly detection, anomalies are rare targets that exhibit distinct spectral signatures from the background. Thus, anomalies are with low probabilities of occurrence in hyperspectral images. In this article, we develop a new technique for hyperspectral anomaly detection that adopts a new information theory perspective, to fully utilize the aforementioned concepts. Our goal is to transform system entropy into quantitative metrics of anomaly conspicuousness of pixels. To do so, two tasks are first completed: first, the construction of occurrence probability of pixels based on the density peak clustering algorithm, and second, the valid system definitions for pixels in specific anomaly detection problems with multiviews. Specifically, three types of systems are separately established by pixel pairs to conform to the definitions of three entropy definitions in information theory, i.e., Shannon entropy, joint entropy, and relative entropy. Then, three individual entropy-based metrics that assess the anomaly conspicuousness are defined. In addition, we design a standard deviation-based ensemble strategy for the integrated representation of the three individual metrics, which considers both logic "OR" and "AND" operations to simultaneously improve the detection rate and reduce the false alarm rate. Our experimental results obtained on two publicly available datasets with anomalies of different sizes and shapes demonstrate the superiority of our newly proposed anomaly detection method.**

***Index Terms*— Anomaly detection, ensemble, entropy, hyperspectral image, information theory, system.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) has the advantage of the integration of image and spectrum with a high spectral resolution, providing the necessary requirements for discovering and extracting land surface types and abnormal targets [1], [2]. HSI has been widely applied to many remote sensing applications, such as classification [3]–[5], target detection [6], [7], and anomaly detection [8], [9]. As a blind signal detection technology, anomaly detection based on spectral feature analysis has been widely used in environmental monitoring, mineral resources exploration, defense security, and other fields [10]–[12] (without prior knowledge of ground objects).

Many methods have been developed for anomaly detection in HSI data. Among them, the Reed–Xiaoli (RX) detector [13] can be considered the cornerstone algorithm. The RX establishes a background Gaussian model for global or local regions and detects anomalies by measuring the Mahalanobis distance from the mean. When the background distribution conforms to the Gaussian distribution, the RX detector can meet the requirements of both computational efficiency and detection accuracy [14]. However, the simple Gaussian model cannot accurately describe the background when the background spectral features are complex. In addition, when there are anomalous pixels in the background, the sensitivity of the background model to anomalies will be reduced, which greatly damages the detection performance [15], [16]. Based on this, the local RX (LRX) is designed, which performs the RX in local windows. However, the size of the local window used for statistical background modeling also affects the accuracy of detection results. As a result, many improvements have been made to the RX to address the aforementioned issues [17]–[24]. For example, the nonlinear kernel RX algorithm [25], [26] added nonlinear features to the image statistics. Schweizer and Moura [27] developed an improved parametric local Gaussian model that uses the Markov random field (MRF) to consider the spatial correlation between adjacent pixels. Moreover, it is suggested that the finite Gaussian mixture models (GMMs) can provide a more accurate characterization of the global background [8] and many GMM-based methods have been developed for anomaly detection [28]–[30]. For example, Veracini *et al.* [28] performed GMM parameters estimation with an extended expectation–maximization algorithm and then detected anomalies through the generalized likelihood ratio test. Some other works adopt nonparametric

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methods for hyperspectral anomaly detection [31]–[33], which does not require any assumptions about the background distribution. For example, in [31], the nonparametric variable-bandwidth kernel density estimator is applied to the log-likelihood-based anomaly detection framework to estimate the background probability density function for detection.

In addition to these methods, the cluster-based anomaly detector (CBAD) [34] and the binary partition tree (BPT)-based detector [35] detect anomalies in multiple homogeneous clusters, which can effectively circumvent the anomaly contamination of the background model. Furthermore, support vector machine (SVM)-based anomaly detection method support vector data descriptions (SVDDs) [36] uses the minimum inclusion hypersphere method to separate anomalies, achieving good detection results. However, for regions where there are multiple background objects, the detection results are usually not stable. In recent years, sparse representation and low-rank decomposition methods have been developed for anomaly detection purposes [37]–[42]. The background joint sparse representation-based anomaly detection (BJSRD) method [43], the low-rank and sparse matrix decomposition-based anomaly detection (LSMAD) algorithm [44], and the collaborative representation-based detector (CRD) [45] are representative examples. The BJSRD uses the local background to extract a sparse background dictionary and then evaluates the pixel anomaly score according to the weight of the pixel to be measured by using the joint sparse representation (JSR). The LSMAD models the background and anomalies by using the low-rank and sparse representation, respectively, according to the principle that background has low-rank attribute and the anomaly is sparse in the global data. The CRD assumes that the background pixels can be represented by their spatial neighborhoods (whereas the anomalies cannot). Different from the above methods, filtering techniques have also been successfully applied for HSI anomaly detection [46]–[48]. For example, the attribute and edge-preserving filtering-based detection method (AED) [46] can provide both superior detection accuracy and time efficiency through few filtering operations. Moreover, Tao *et al.* [49] proposed a novel detection method using the fractional Fourier transform to obtain discriminant features between targets and background, which also provided excellent detection performance. Recently, several autoencoder (AE)-based methods have been developed to extract spatial–spectral feature matrices for anomaly detection purposes [50]–[53], achieving satisfactory detection results.

At this point, it is important to mention that—up to now—there is no clear and authoritative definition of "abnormal target" in HSI anomaly detection. It is generally accepted that anomalies in HSI data usually refer to spectral anomalies. If the target features are obviously different from the background information, they can be defined as a possible abnormal target. As a result, different from the traditional anomaly detection methods based on high spatial resolution remote sensing images, all the aforementioned methods conducting anomaly detection are based on statistical modeling or representation theory to evaluate the spectral difference between anomalous and background pixels [54].

In addition, the property that the anomaly pixels are rare and with low probabilities of occurrence in the image is also the consensus in anomaly detection. We think that this is a useful property based on the following reason. Once the occurrence probability of each pixel is obtained, we can distinguish background and anomaly according to the principle that the occurrence probability of an anomaly pixel is low and that of a background pixel is high. Motivated by this, we propose a new anomaly detection method based on information entropy analysis to fully consider the aforementioned property. According to the definition, information entropy is a quantitative index of information content of a system, calculated with the probabilities of events. For the two factors (system and event probability) needed, if we generate appropriate and valid system definitions for pixels with the known occurrence probability, entropy can be transformed into a metric of the anomaly conspicuousness.

To do so, we define a system for each test pixel. The system is constructed by the test pixel and each of its neighboring pixels (the neighborhood considered is the local window centered on the test pixel). In this system, two pixels can be defined as two events belonging to a random variable or as two independent random variables (each variable contains two events). The key point is to calculate the entropy of the system through the probability of the events in the system. We assume that most of the pixels in the neighborhood are background pixels and then jointly consider the entropy of the system constructed by the test pixel and each of its neighboring pixels to reflect whether the test sample belongs to the background or anomaly. Specifically, three kinds of entropy definitions in information theory are considered, i.e., the Shannon entropy (SE), joint entropy (JE), and relative entropy (RE). Accordingly, three types of systems are separately constructed for each test pixel. Then, three individual entropy-based metrics are used to assess anomaly conspicuousness by jointly calculating the system entropy of pixel systems in local windows. In order to achieve more robust detection performance, we design a standard deviation-based ensemble strategy for the integrated representation of the three individual metrics, which considers both "OR" and "AND" logic operation to simultaneously improve the detection rate and reduce the false alarm rate. Experimental results are conducted on the Airport-Beach-Urban (ABU) and Viareggio datasets, containing different sizes of targets over different scenes, demonstrating the superiority of our newly proposed anomaly detection method when compared to other widely used competitors.

The remainder of this article is organized as follows. Section II describes some related works. Section III describes the proposed hyperspectral anomaly detection method in detail. Section IV provides the experimental setup and presents our extensive evaluation. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

In this section, we briefly revisit some related works, including the density peak clustering (DPC) algorithm [55] and some basic concepts about entropy in information theory.

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A. DPC Algorithm

The density peak clustering (DPC) algorithm proposed by Rodriguez and Laio [55] is a data-driven adaptive clustering algorithm. For each pixel $\mathbf{x}_i$ in the dataset, the distance d_{ij} between $\mathbf{x}_i$ and other pixels $\mathbf{x}_j$ is first computed as follows:

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2 \tag{1}$$

where $\|\cdot\|_2$ denotes the $\mathbf{L}_2$ norm. The value of d_{ij} measures the similarity between two pixels $\mathbf{x}_i$ and $\mathbf{x}_j$. The smaller the distance, the more similar the two pixels are, and vice versa.

For $\mathbf{x}_i$, the density and distance parameters of $\mathbf{x}_i$ are defined according to the d_{ij} set. To reduce the error of density estimation caused by the number of pixels around the pixel to be measured, the Gaussian radial basis function (GBF) is used to define the density [56]. According to the definition of GBF, the density ρ_i of pixel $\mathbf{x}_i$ is defined as

$$\rho_i = \sum_j \exp\left[\left(-\frac{d_{ij}}{d_c}\right)^2\right] \tag{2}$$

where d_c is the specified cutoff distance that determines the radius of the scale of similarity measurement. The value of density ρ_i is determined by the number of pixels with similar spectral features to pixel $\mathbf{x}_i$. The higher the density ρ_i is, the more similar pixels to $\mathbf{x}_i$ are. If pixel $\mathbf{x}_i$ is taken as the center, the set size of similar pixels around it will be large in the distance space and vice versa.

For any pixel $\mathbf{x}_i$ in the dataset, according to its density ρ_i and densities of other pixels ρ_j, the distance parameter is defined as

$$\delta_i = \begin{cases} \min\limits_{j:\rho_j>\rho_i}(d_{ij}), & \exists\ \rho_i < \rho_j \\ \max\limits_{j}(d_{ij}), & \text{others} \end{cases} \tag{3}$$

where δ_i reflects the significance of pixel $\mathbf{x}_i$ as the cluster center. When the density of pixel $\mathbf{x}_i$ is the maximum, δ_i takes the maximum value in the set of d_{ij}. On the contrary, if the density of pixel $\mathbf{x}_j$ is larger than that of pixel $\mathbf{x}_i$, then δ_i takes the minimum distance between $\mathbf{x}_j$ and $\mathbf{x}_i$. In this article, we adopt the density definition in DPC to evaluate the occurrence probability for each pixel.

B. Entropy in Information Theory

Entropy is a very important concept and physical quantity in physics but also widely used in many fields. For the sake of distinction, the entropy in statistical physics is usually called physical entropy, and the entropy in information theory is called information entropy. In 1-D random systems, information entropy is also called SE, which measures the amount of information (that is, the uncertainty of the system). In 2-D random systems, many concepts are derived from information entropy, such as JE and RE. JE is the representation of information entropy in a 2-D system, which represents the uncertainty of the system in which two events occur simultaneously. RE is equivalent to the difference of information entropy of two probability distributions in two distribution systems. In this article, the SE, JE, and RE are transformed to metrics that evaluate the anomaly conspicuousness from different views and integrate them to an ensemble entropy metric-based detector (EEMD) that is described in Section III.

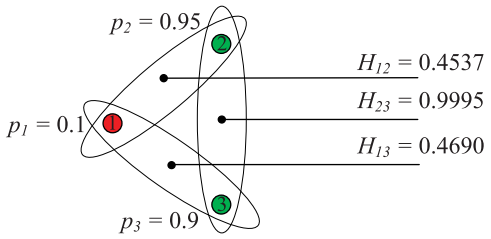


Fig. 1. Simple illustration of the motivation. The green and red points represent the background and anomaly pixels, respectively. P_i $(i = 1, 2, 3)$ is the occurrence probability. The entropy H is calculated by the SE, i.e., $H_{ij} = -\sum_{k=i,j} p(\mathbf{x}_k)\log\ p(\mathbf{x}_k)$, where $\{p(\mathbf{x}_k)\}_{k=i,j} = \{(P_k/(P_i + P_j))\}_{k=i,j}$.

III. Ensemble Entropy Metric-Based Detector (EEMD)

A. Motivation

In hyperspectral anomaly detection, the definition of the anomalies is unique. Anomalies are rare targets that exhibit distinct spectral signatures from the background. Thus, anomalies are with low probabilities of occurrence in hyperspectral images. This is a useful property for anomaly detection. To take advantage of this property, the occurrence probability of each pixel in the image needs to be determined first. Furthermore, in order to effectively use the occurrence probability of pixels to distinguish background and anomaly, the information entropy is introduced to anomaly detection, which is defined as a quantitative index of information content of a system, calculated with the probabilities of events. Our goal is to transform system entropy into quantitative metrics of anomaly conspicuousness with the occurrence probability, for the identification of background and anomaly. Fig. 1 shows a simple example to illustrate the feasibility of this idea. In Fig. 1, the two green points represent two background pixels with the occurrence probabilities of 0.95 and 0.9, and the red point represents an anomaly pixel with the occurrence probability of 0.1. As can be seen, the entropy between the anomaly pixel and the background pixel is significantly less than that between two background pixels. Therefore, once the occurrence probability of each pixel is obtained, we can distinguish background and anomaly pixels.

However, the calculation of entropy in Fig. 1 is only a mechanical application of the formula, and the result lacks interpretability and concrete meaning. As a result, valid system definitions for individual pixels (or between pixels) are required for the explanation of our specific anomaly detection problem. In short, to achieve our goal, two tasks need to be completed: first, the construction of occurrence probability of pixels, and second, the valid system definitions for specific problem. In addition, to obtain more robust detection performance, three kinds of entropy definitions in information theory are considered, i.e., the SE, JE, and RE, with multiviews for system construction.

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B. Construction of Occurrence Probability

For the realization of the three entropy-based metrics in subsequent sections, we need first to construct the occurrence probability for each pixel in the whole image. In general, the occurrence probability of background pixels and anomaly pixels can be represented by the proportion of their number in the image. However, in anomaly detection, the ground truth is unavailable in the detection stage, that is, we cannot obtain the occurrence probability in advance. In this article, we use the density obtained by the DPC algorithm to represent occurrence probability for the following reasons. First, due to the outlier characteristics in features and the rarity in quantity, the anomaly pixels exhibit low density, while the background pixels have the opposite. Second, the DPC algorithm can adaptively calculate the density driven by the attributes of data itself according to the similarity between data in a certain measure. Third, according to the definition of DPC as described in Section II-A, the higher the density of a certain pixel is, the more similar pixels to this pixel are, which just corresponds to that background pixels have high density, and thus, the density value is highly positively correlated with the probability that the pixel is background. Finally, the DPC algorithm has been widely and successfully applied in hyperspectral image processing [57]–[59], which shows the applicability of the algorithm for hyperspectral data.

Let $\mathbf{X} \in \mathbb{R}^{I\times D}$ be a hyperspectral image with I pixels with D spectral bands. Since the DPC process requires computing the distance matrix among pixels, processing the entire image at once requires a large amount of memory and a very high time consumption. Therefore, in our implementation, the HSI $\mathbf{X}$ is first divided into a certain number of nonoverlapping blocks with size $w_p \times w_p$ and the DPC is then conducted on each block in sequence. This process can be simply described as follows:

$$\boldsymbol{\rho} = \mathrm{DPC}(\mathbf{X}, w_p, r_c) \tag{4}$$

where DPC$(\cdot)$ is the DPC operation, w_p is the block size and r_c controls the cutoff distance d_c. To be specific, let $\mathbf{D}$ be the distance matrix for the block. We extract the upper triangular elements of $\mathbf{D}$ and sort them as vector $\mathbf{d}$ in ascending order. Then, we have $d_c = \mathbf{d}([T(T-1)\cdot r_c\%])$, where $T = w_p \times w_p$ and $r_c\%$ is the percentage, which determines the position of d_c in $\mathbf{d}$. $\boldsymbol{\rho} = \{\rho_i\}_{i=1}^{I} \in \mathbb{R}^{I\times 1}$ (values range from 0 to 1) is the output density vector composed of each pixel density, namely the occurrence probability for each pixel. In this article, the initial background probability refers to the probability that a certain pixel is a background pixel. It is obvious that the occurrence probability is proportional to the background probability of pixels. To some extent, the initial background probability can be represented directly by the occurrence probability.

C. Realization of Entropy-Based Metrics

In order to obtain the anomaly score (anomaly conspicuousness) with entropy-based metrics for each pixel under test (PUT), a sliding window operation is adopted. For each PUT $\mathbf{x} \in \mathbb{R}^{D\times 1}$, the surrounding pixels within the window (of the size of $w_e \times w_e$), centered at the PUT $\mathbf{x}$, are used to jointly calculate the anomaly score of $\mathbf{x}$ by evaluating the similarity between them, based on the assumption that most of the surrounding pixels are normal (background) pixels. It is worth mentioning that the use of an inner and outer window is suggested in many works [36], [45] to achieve a more robust detection result. In such a strategy, only the samples inside the outer window and outside the inner window are collected, while the pixels inside the inner window are treated as anomalies similar to the central PUT, which may interfere with the reconstruction error. However, our experiments show that, when the window size w_e is large enough, the effect of whether using a hollow (inner and outer) window strategy or not is negligible in our method. Consequently, in order to simplify the algorithm and avoid additional parameter settings, we adopt a sliding window strategy with a single window. In addition, for each PUT $\mathbf{x}_i$ $(i = 1, 2, \ldots, I)$ with the initial background probability (occurrence probability) ρ_i, the surrounding pixels are given by $\{\mathbf{x}_j\}_{j=1}^{w_e\times w_e-1}$ with the corresponding initial background probability $\{\rho_j\}_{j=1}^{w_e\times w_e-1}$.

After discussing the above aspects, we now design three different entropy metrics based on three kinds of entropy concepts in information theory and then integrate them into an ensemble metric for anomaly detection. Fig. 2 shows the schematic of the ensemble entropy metric-based anomaly detector.

1) SE-Based Metric: The SE describes the indeterminacy (i.e., the average amount of information source) of a random variable. In order to implement this concept, we first define a random variable (single-variable system) and then make a logical derivation as follows. Consider the random variable

$$\begin{aligned}\xi &= \text{“Which of } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ is closer to } \mu\text{”}\\ &= \{\mathbf{x}_i, \mathbf{x}_j\}\end{aligned} \tag{5}$$

where μ is the mean of the surrounding pixel set $\{\mathbf{x}_j\}_{j=1}^{w_e\times w_e-1}$ and is viewed as a background pixel vector because the surrounding pixels are assumed to be background pixels. Therefore, a PUT close to μ is equivalent to the fact that the PUT tends to be background, that is, the probability of the PUT being close to μ is equal to that of the PUT being background. According to this, we can obtain

$$p(\xi = \mathbf{x}_k) = p(\text{“}\mathbf{x}_k \text{ is background”}) = \rho_k, \quad (k = i, j). \tag{6}$$

Then, the SE of the random variable ξ can be written as follows:

$$H_{\mathrm{SE}}{}^{ij} = -\sum_{k=i,j} p(\mathbf{x}_k)\log\ p(\mathbf{x}_k) \tag{7}$$

where $\{p(\mathbf{x}_k)\}_{k=i,j} = \{\rho_k/(\rho_i+\rho_j)\}_{k=i,j}$ is the normalized background probability between $\mathbf{x}_i$ and $\mathbf{x}_j$. This step is to ensure that the sum of the probabilities of the two is equal to 1 to match the definition of SE.

According to the maximum entropy theorem, the more uniform the probability distribution of the random variable is, the larger the SE is. When the probability of each event is equal, the maximum SE will be obtained, that is, the higher the similarity between ρ_i and ρ_j, the larger the $H_{SE}{}^{ij}$. Now, we

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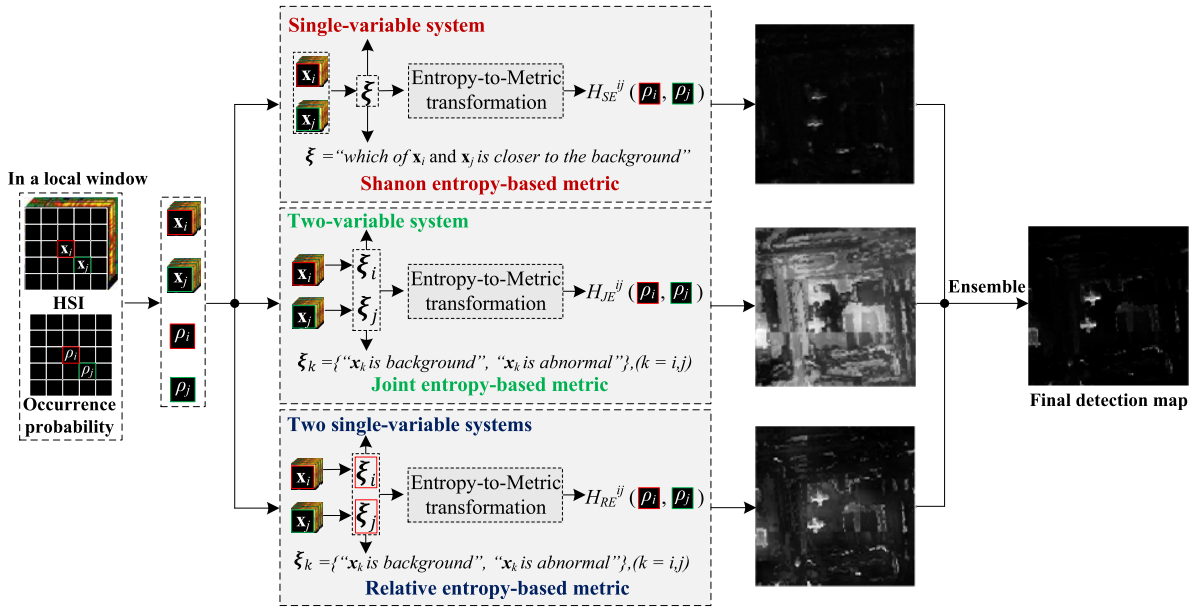


Fig. 2. Schematic of the ensemble entropy metric-based anomaly detector.

can reinterpret (7) as a metric that measures the similarity between ρ_i and ρ_j (equivalent to the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$). Thus, the similarity between the PUT $\mathbf{x}_i$ and the surrounding pixels (background) set $\{\mathbf{x}_j\}_{j=1}^{w_e\times w_e-1}$ is defined by

$$H_{\mathrm{SE}}{}^{i} = \frac{1}{w_e \times w_e - 1} \sum_{j=1}^{w_e\times w_e-1} H_{\mathrm{SE}}{}^{ij}. \tag{8}$$

2) JE-Based Metric: Different from the SE-based metric (SED) where $\mathbf{x}_i$ and $\mathbf{x}_j$ are regarded as two random events of a random variable, here, $\mathbf{x}_i$ and $\mathbf{x}_j$ are defined as two independent random variables in a two-variable system, and their similarity is described by the degree of random system chaos. Specifically, two independent random variables are defined as

$$\begin{aligned}\xi_k &= \text{“Is } \mathbf{x}_k \text{ background or abnormal”} \\ &= \{\text{“}\mathbf{x}_k \text{ is background”, “}\mathbf{x}_k \text{ is abnormal”}\} \\ &\quad (k = i, j).\end{aligned} \tag{9}$$

According to the initial background probability $\boldsymbol{\rho}$, we can have

$$p(\text{“}\mathbf{x}_k \text{ is background”}) = \rho_k, \quad (k = i, j) \tag{10}$$

and

$$p(\text{“}\mathbf{x}_k \text{ is abnormal”}) = 1 - \rho_k, \quad (k = i, j). \tag{11}$$

For the two-variable random system composed by the two random variables ξ_i and ξ_j, a joint probability distribution needs to be constructed. Since our method starts with the spectral and density properties of the pixels without considering the spatial correlation among them, whether a pixel is background or abnormal is irrelevant to other pixels. Hence, ξ_i and ξ_j are pairwise independent variables. Consequently, with the marginal distribution from Formulas (10) and (11), we can easily compute the joint probability distribution of ξ_i and ξ_j, as shown in Fig. 3. For convenience, we use ξ_{k1} and ξ_{k2} to refer to the events “$\mathbf{x}_k$ is background” and “$\mathbf{x}_k$ is abnormal” $(k = i, j)$, respectively, in Fig. 3.

ξ_j \ ξ_i	ξ_{i1}	ξ_{i2}	marginal distribution of ξ_j
ξ_{j1}	$p(\xi_{i1},\xi_{j1}) = \rho_i \times \rho_j$	$p(\xi_{i2},\xi_{j1}) = \rho_j - p(\xi_{i1},\xi_{j1}) = \rho_j \times (1-\rho_i)$	ρ_j
ξ_{j2}	$p(\xi_{i1},\xi_{j2}) = \rho_i - p(\xi_{i1},\xi_{j1}) = \rho_i \times (1-\rho_j)$	$p(\xi_{i2},\xi_{j2}) = (1-\rho_i) \times (1-\rho_j)$	$1-\rho_j$
marginal distribution of ξ_i	ρ_i	$1-\rho_i$	

Fig. 3. Construction of the joint probability distribution. ξ_{k1} and ξ_{k2} refer to the events “$\mathbf{x}_k$ is background” and “$\mathbf{x}_k$ is abnormal” $(k = i, j)$, respectively.

Now, the JE of the random system composed by ξ_i and ξ_j is expressed as follows:

$$H_{\mathrm{JE}}{}^{ij} = -\sum_{u\in\xi_i}\sum_{v\in\xi_j} p(u,v)\log\ p(u,v) \tag{12}$$

where $\xi_i = \{\xi_{i1}, \xi_{i2}\}$ and $\xi_j = \{\xi_{j1}, \xi_{j2}\}$. The maximum entropy theorem is also applicable to JE. This means that a more uniform joint probability distribution $p(u, v)$ leads to a larger JE $H_{\mathrm{JE}}{}^{ij}$. According to Fig. 3, we know that the uniformity of joint distribution is positively correlated with that of marginal distribution. In addition, the more uniform the marginal distribution means that both ρ_i and ρ_j tend to be close to 0.5 (i.e., the density of $\mathbf{x}_i$ and $\mathbf{x}_j$ tends to be the same, and $\mathbf{x}_i$ and $\mathbf{x}_j$ are similar. As a result, (12) can also be viewed as a metric evaluating the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$, and the

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similarity between $\mathbf{x}_i$ and $\{\mathbf{x}_j\}_{j=1}^{w_e\times w_e-1}$ is defined as follows:

$$H_{\mathrm{JE}}{}^{i} = \frac{1}{w_e \times w_e - 1} \sum_{j=1}^{w_e\times w_e-1} H_{\mathrm{JE}}{}^{ij}. \tag{13}$$

3) RE-Based Metric: RE is a measure of the difference between two probability distributions P and Q. The divergence is used to measure the average additional number of bits required to encode samples from P using Q-based coding. Typically, P is the true distribution of data, and Q is the theoretical distribution, model distribution, or approximate distribution of P, that is to say, RE can measure the distance between two random distributions. When two random distributions are the same, their RE is zero. When the difference between two random distributions increases, their RE will also increase.

In Section III-C2, we have defined two random variables (viewed as two single-variable systems here) ξ_i and ξ_j and given their probability distribution. Then, the RE between ξ_i and ξ_j is given by

$$H_{\mathrm{RE}}{}^{ij} = -\sum_{t=1}^{2} p(\xi_{jt}) \log \frac{p(\xi_{jt})}{p(\xi_{it})}. \tag{14}$$

According to simple reasoning, the distance between the probability distributions of ξ_i and ξ_j is equivalent to the distance between the normal probabilities of $\mathbf{x}_i$ and $\mathbf{x}_j$ (namely, the density of $\mathbf{x}_i$ and $\mathbf{x}_j$). Consequently, (14) can be transformed into a distance metric of two pixels $\mathbf{x}_i$ and $\mathbf{x}_j$. As a result, we have

$$H_{\mathrm{JE}}{}^{i} = \frac{1}{w_e \times w_e - 1} \sum_{j=1}^{w_e\times w_e-1} H_{\mathrm{RE}}{}^{ij} \tag{15}$$

which is the distance between the PUT $\mathbf{x}_i$ and its surrounding pixels $\{\mathbf{x}_j\}_{j=1}^{w_e\times w_e-1}$.

D. Ensemble Entropy Metric-Based Detector for Anomaly Detection

The three aforementioned entropy-based metrics transfer three different entropy concepts into the evaluation of the similarity of the PUTs and their surrounding pixels by constructing random variables and systems and deriving their internal relations, from different perspectives.

1) The SED takes two pixels as two random events in a random variable (single-variable system) and converts their initial background probabilities into event probabilities. According to the maximum entropy theorem, the probability distribution uniformity of this random variable is measured by information entropy.
2) The JE-based metric (JED) defines two random variables for two pixels, calculates the JE of the random system composed of these two random variables, and then converts the JE into the measure of the initial background probability similarity.
3) The RE-based metric (RED) calculates the distance between the probability distributions of these two random variables directly and takes it as the difference of the initial background probability between pixels.

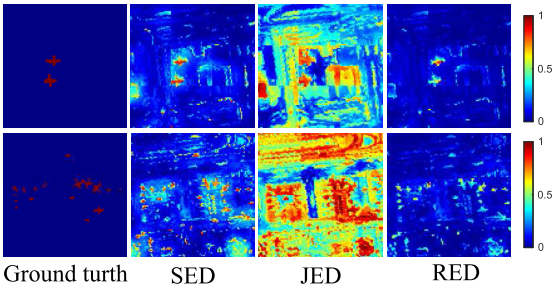


Fig. 4. Anomaly map obtained by the SED, JED, and RED, respectively. Taking ABU-airport (b) and (c) for examples.

Unfortunately, as shown in Fig. 4, the detection results of SED can highlight some anomalies but with several false detections; the JED can detect the anomaly completely, but it has little background suppression ability; and the RED has better background suppression, but the anomaly emphasis is worse than SED. Therefore, each metric has its drawbacks, leading to detection performance less than ideal when used alone. This section integrates the three individual metrics as an EEMD to obtain more robust detection results. Let $\mathbf{H}_{\mathrm{SE}}$ refer to the matrix consisting of the element $H_{\mathrm{SE}}{}^{i}$ of all pixels, likewise $\mathbf{H}_{\mathrm{JE}}$ and $\mathbf{H}_{\mathrm{RE}}$. After normalizing $\mathbf{H}_{\mathrm{SE}}$, $\mathbf{H}_{\mathrm{JE}}$, and $\mathbf{H}_{\mathrm{RE}}$ to 0∼1, we define ${}^{O}\mathbf{H}_{\mathrm{SE}} = \mathbf{E} - \mathbf{H}_{\mathrm{SE}}$, ${}^{O}\mathbf{H}_{\mathrm{JE}} = \mathbf{E} - \mathbf{H}_{\mathrm{JE}}$, and ${}^{O}\mathbf{H}_{\mathrm{RE}} = \mathbf{H}_{\mathrm{RE}}$ as the anomaly scores obtained by the SED, JED, and RED, respectively, where $\mathbf{E}$ is a matrix with element values of 1 and the same size as $\mathbf{H}_{\mathrm{SE}}$, $\mathbf{H}_{\mathrm{JE}}$, and $\mathbf{H}_{\mathrm{RE}}$.

We first adopt the sum rule for the integration of the individual metrics as follows:

$${}^{O}\mathbf{OR} = \frac{1}{3}\left({}^{O}\mathbf{H}_{\mathrm{SE}} + {}^{O}\mathbf{H}_{\mathrm{JE}} + {}^{O}\mathbf{H}_{\mathrm{RE}}\right). \tag{16}$$

This operation is a simple bagging model averaging multiple anomaly scores, which can also be viewed as a soft max pooling over the anomaly scores. The sum rule here implements the logical "OR" operation, namely a disjunction of all anomalies of the individual metrics. It can effectively reduce the missed detection rate, that is, improve the detection rate.

Second, the product rule is also considered as follows:

$${}^{O}\mathbf{AND} = {}^{O}\mathbf{H}_{\mathrm{SE}} \times {}^{O}\mathbf{H}_{\mathrm{JE}} \times {}^{O}\mathbf{H}_{\mathrm{RE}} \tag{17}$$

which can be understood as the logical "AND" operation whose result is true only when all conditions are true. It implements conjunction of all anomalies of the individual metrics, which can effectively reduce the false alarm rate, based on that a PUT is an anomaly only when it is judged to be abnormal by each of the individual metrics.

We then generate ensemble entropy-based metric by combining the sum- and product-rule integrations of the three individual metrics, which is formulated as

$${}^{O}\mathbf{EEMD} = \mathcal{W} \cdot *{}^{O}\mathbf{OR}^{3} + (\mathbf{E} - \mathcal{W}) \cdot *{}^{O}\mathbf{AND} \tag{18}$$

where ${}^{O}\mathbf{EEMD}$ is the final anomaly map and $\mathcal{W}$ is a weight map (where each element range from 0 to 1) controlling the contribution of the sum and product rules. Furthermore, we

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Fig. 5. Binary images with the same histogram, but with different spatial structures.

propose an image standard deviation-based strategy to automatically determine the weight map $\mathcal{W}$

$$\tilde{\mathcal{W}} = \arg\min_{\mathcal{W}} \ \mathcal{STD}({}^{O}\mathbf{EEMD}) \tag{19}$$

where $\mathcal{STD}({}^{O}\mathbf{EEMD})$ calculates the image standard deviation of the ${}^{O}\mathbf{EEMD}$ that can be understood as a grayscale image. This strategy is motivated by: in an ideal anomaly map, the anomaly regions would be assigned higher values, and the background of the image would be largely suppressed. Therefore, we expect values in the histogram of the anomaly map to be clustered around a specific value. Thus, according to the definition of the standard deviation of the images, the standard deviation of a desirable anomaly map will be very small.

Conventional image standard deviation is determined by the histogram of an image, without considering the spatial structure information. As can be seen from Fig. 5 [60], even though the spatial structure becomes more and more chaotic, images exhibit the same histograms (white and black account for half of the image, respectively) and therefore have the same standard deviation values. Obviously, it is expected to obtain an anomaly map with a low degree of chaos in anomaly detection. As a result, we take the spatial structure information into account. Inspired by [60], we integrate two simple strategies to make use of the spatial geometric information. First, a simple improved definition of standard deviation is provided, where the intensity of each pixel in the anomaly map is required to depend on the values of its neighbors. This objective can be achieved with many spatial filters. In this article, the anomaly map is filtered by using a low-pass Gaussian kernel, and then, the conventional standard deviation is computed on the filtered anomaly map. Consequently, the new standard deviation is defined as: $\mathcal{STD}_{2-\mathrm{D}}(\mathcal{A}) = \mathcal{STD}(g \star \mathcal{A})$, where g is a low-pass Gaussian kernel. Second, we employ a border avoidance strategy [61] to avoid anomaly maps with a strong border response. The border effect factor α is defined as: given an anomaly map $\mathcal{A}$, $\alpha = \sum\sum \mathcal{G}(n,m) \cdot \mathcal{N}(A(n,m))$, where $\mathcal{G}$ is a Gaussian mask of the same size as $\mathcal{A}$, and $\sum\sum \mathcal{G}(n,m) = 1$. $\mathcal{N}(\cdot)$ is used to normalize $\mathcal{A}$ so that the summation of all the pixel values is 1. Consequently, $\tilde{\mathcal{W}}$ in (19) is revised as follows:

$$\tilde{\mathcal{W}} = \arg\min_{\mathcal{W}} \ \alpha^{-1}\mathcal{STD}_{2-\mathrm{D}}\left({}^{O}\mathbf{EEMD}\right). \tag{20}$$

In this article, the interior point algorithm is adopted for optimizing the weight map $\mathcal{W}$. However, if we directly optimize all the pixels of the whole weight map in (20), the number of independent variables (equal to the number of pixels) to be optimized will be very large, leading to expensive computing consumption and memory requirements. For example, a 100×100 anomaly map leads to 10 000 independent variables. We use the blocking strategy as shown in Fig. 6 to address this problem, where the anomaly map is divided into several $N \times N$ blocks, and the pixels in the same block are given the same weight value. In our implementation, we set $N = 10$ for the ABU dataset and $N = 25$ for the Viareggio dataset, and these datasets will be described in the experimental part. Moreover, the edge padding can be used to deal with the case that the edge length of the image cannot be divided by N. Based on this, $\mathcal{W}$ is optimized, and then, the final anomaly map ${}^{O}\mathbf{EEMD}$ is obtained for anomaly detection purposes. The proposed method is summarized in Algorithm 1.

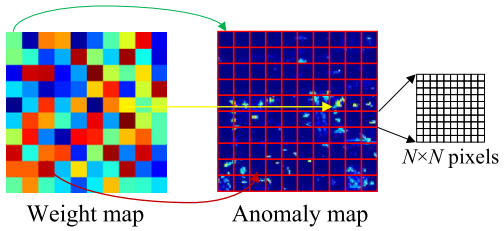


Fig. 6. Sketch map for the blocking strategy.

Algorithm 1 EEMD

Inputs: The HSI $\mathbf{X}$; Parameters w_p, r_c and w_e
Output: Anomaly map ${}^{O}\mathbf{EEMD}$ of $\mathbf{X}$.

1: Construct the occurrence probability vector $\boldsymbol{\rho}$ of $\mathbf{X}$ according to (4)
2: Compute the anomaly map by the SE-based metric, ${}^{O}\mathbf{H}_{SE}$, according to (7) and (8)
3: Compute the anomaly map by the JE-based metric, ${}^{O}\mathbf{H}_{JE}$, according to (12) and (13)
4: Compute the anomaly map by the RE-based metric, ${}^{O}\mathbf{H}_{RE}$, according to (14) and (15)
5: Obtain the logical "OR" detection map, ${}^{O}\mathbf{OR}$, by (16)
6: Obtain the logical "AND" detection map, ${}^{O}\mathbf{AND}$, by (17)
7: Optimize the weight map $\mathcal{W}$ according to (20)
8: Combine ${}^{O}\mathbf{OR}$ and ${}^{O}\mathbf{AND}$ by (18) to obtain the final Anomaly map ${}^{O}\mathbf{EEMD}$
9: **return** ${}^{O}\mathbf{EEMD}$

IV. Experiments

A. Hyperspectral Datasets

The first dataset used in this article is the ABU dataset, which includes an Airport scene (four images), a Beach scene (four images), and an Urban scene (five images). This dataset is available online.[1] The sample images and reference maps are separately shown in the first and second columns of Figs. 14–16. The images with the size of 100×100 or 150×150 pixels are manually extracted from larger images available on the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) website.[2] The anomalies are the most abnormal targets determined by human vision. The ground truth of anomalies was manually labeled with the Environment

[1] http://xudongkang.weebly.com/
[2] http://aviris.jpl.nasa.gov/

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TABLE I
DETAILED FEATURES OF THE ABU DATASET

Images		Captured place	Resolution	Length×Width×Bands
ABU-airport	(a)	Los Angeles	7.1m	100×100×205
	(b)	Los Angeles	7.1m	100×100×205
	(c)	Los Angeles	7.1m	100×100×205
	(d)	Gulfport	3.4m	100×100×191
ABU-beach	(a)	Cat Island	17.2m	150×150×188
	(b)	San Diego	7.5m	100×100×193
	(c)	Bay Chamoagne	4.4m	100×100×188
	(d)	Pavia	1.3m	150×150×102
ABU-urban	(a)	Texas Coast	17.2m	100×100×204
	(b)	Texas Coast	17.2m	100×100×207
	(c)	Gainesville	3.5m	100×100×191
	(d)	Los Angeles	7.1m	100×100×205
	(e)	Los Angeles	7.1m	100×100×205

for Visualizing Images (ENVI) software. All images were captured by the AVIRIS sensor except the Beach-(4) image, which was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. More detailed features of these images can be found in Table I [46].

The second dataset is the Viareggio dataset, which is available online.[3] This dataset was collected during the "Viareggio 2013 trial" hyperspectral data collection campaign [62], in Viareggio, Italy. It has a size of 375 × 450 pixels with a 0.6-m spatial resolution and contains 511 spectral bands with a 2.3-nm spectral resolution. In the scene, three vehicles, four panels, and two reference calibration tarps with 135 anomalous pixels are labeled anomalies, as shown in Fig. 19.

B. Evaluation Criteria

The receiver operating characteristic (ROC) curves [63] and the area under the curves (AUC) are used to conduct an exhaustive evaluation in this article. The ROC curves can be generated by any two of the true positive rate P_d, false positive rate P_f, and threshold τ. The ROC curve of (P_d, P_f) indicates the tradeoff between P_d and P_f. The ROC curve of (P_f, τ) is a metric of background suppression. The AUC scores are the quantifications of ROC curves, which can describe the detector performance more explicitly. A higher AUC score of (P_d, P_f) means better detection performance, and a lower AUC score of (P_f, τ) means fewer background pixels that are mistakenly detected as anomalies.

C. Evaluating the Proposed EEMD

In this section, the effectiveness of the strategy of EEMD is evaluated by comparing it to other detectors using individual metrics. We use SED, JED, and RED to refer to the detectors using the SED, JED, and RED, respectively. In addition, the detection results obtained by directly using the initial background probability are also considered for comparison, which is abbreviated as INPD.

[3]http://rsipg.dii.unipi.it/

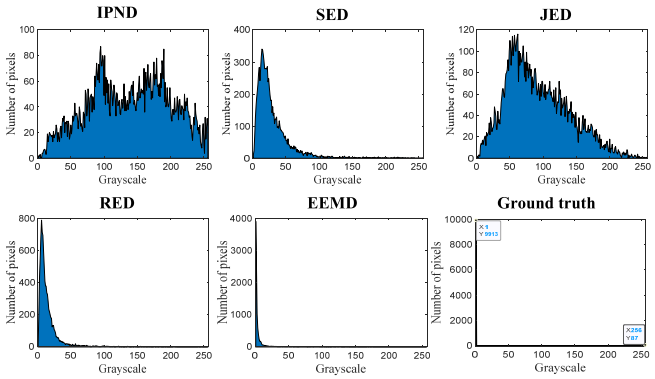


Fig. 7. Detection results and their corresponding gray histograms obtained by the INPD, SED, JED, RED, and EEMD on the ABU-airport-(b) image.

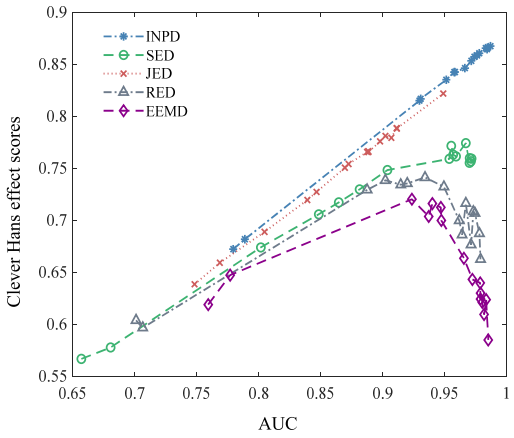


Fig. 8. Curves of AUC score and CHE score for each of the INPD, SED, JED, RED, and EEMD on the ABU-airport-(b) image.

1) Grayscale Distribution Evaluation: Fig. 7 shows the detection result maps and their corresponding gray histograms obtained by different detectors (experiments are conducted on colorblueABU-airport-(b) image). As it can be observed intuitively, each of the EEMD, SED, JED, and RED provides a great improvement of the histogram distributions compared to the INPD, and the EEMD performs best. We can observe that the histogram of the EEMD is much closer to the ground truth than other detectors (especially the INPD). This indicates not only the feasibility of individual entropy-based metrics in anomaly detection but also the effectiveness of the proposed ensemble strategy.

2) "Clever Hans" Effect: "Clever Hans" refers to a horse that can perform arithmetic operations. Later research has proved that it does not really know arithmetic but relies on the unconscious hints given by the trainer or observer to give the correct answer [64]. In machine learning (ML), the "Clever Hans" effect (CHE) shows that an ML model may not really learn the relevant knowledge to solve specific problems and undermines the generalization capability of the

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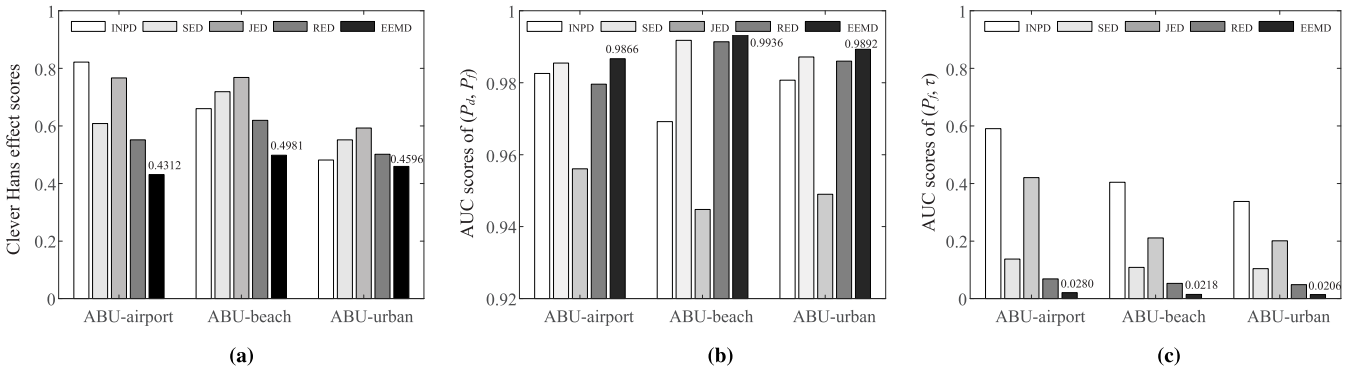


Fig. 9. (a) CHE score, (b) AUC score of (P_d, P_f), and (c) AUC score of (P_f, τ) for the INPD, SED, JED, RED, and EEMD, averaged over all images of each scene of the ABU dataset.

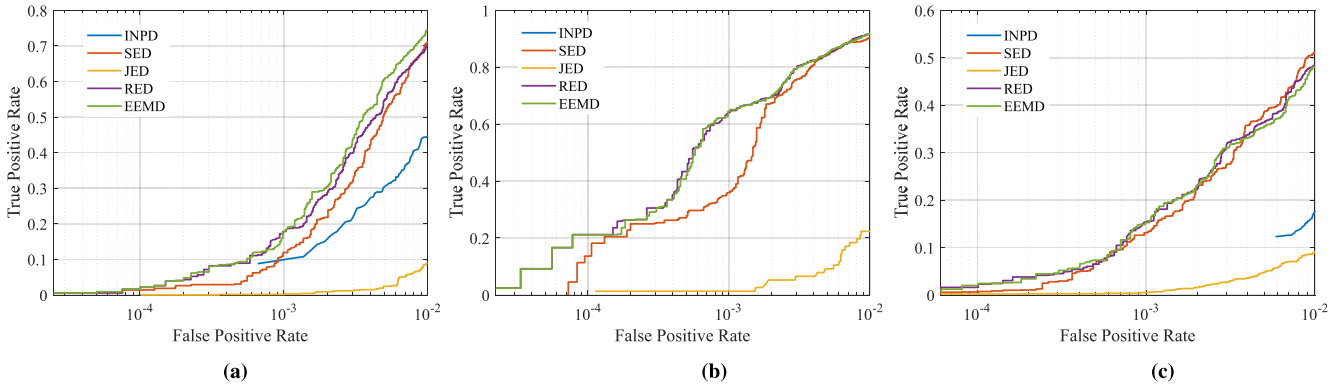


Fig. 10. ROC curves for the INPD, SED, JED, RED, and EEMD, averaged over all images of each scene of the ABU dataset. (a) ABU-airport scene. (b) ABU-beach scene. (c) ABU-urban scene.

model, which has been frequently observed in information retrieval [65] and supervised learning [66]. Very recently, Kauffmann *et al.* [64] revealed the CHE in anomaly detection for the very first time and believed that this effect in anomaly detection was not caused by data but by the structure of the detector themselves. Inspired by this, we further assess the effectiveness of the integration strategy and the generalization ability of the proposed EEMD method by analyzing the CHE of detectors.

In anomaly detection, the CHE can be intuitively expressed as high detection accuracies but poor pixelwise explanation results (namely the detection result maps as shown in Figs. 14–16). In this case, we can assume that part of the accuracy is the result of the CHE. Similar to [64], we define the CHE score by the difference between detection accuracy and explanation accuracy. The detection accuracy is the AUC score of (P_d, P_f) and the explanation accuracy is computed as the cosine similarity between the pixelwise explanation (namely the pixelwise anomaly score) and the ground truth. We then have

$$\text{CHE} = \text{detection accuracy} - \text{explanation accuracy}. \quad (21)$$

The lower the CHE, the stronger the true detection capability and generalization capability of the detectors. Fig. 8 shows the changes of the CHE with the AUC scores of (P_d, P_f) for each detector, where different scores are obtained under different parameter settings. Under the same AUC score of (P_d, P_f), the EEMD always seems to have the lowest CHE. For the INPD and JED, AUC score of (P_d, P_f) and CHE score are very close to positive linear correlation, meaning

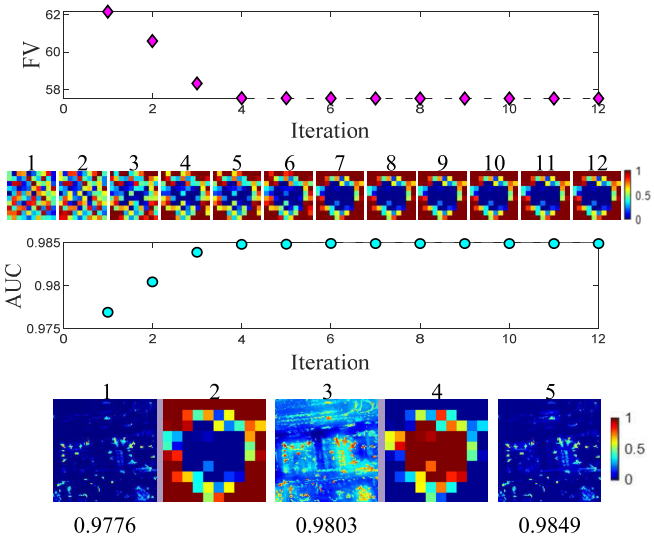


Fig. 11. (Row 1) Iteration process when optimizing $\mathcal{W}$. (Row 2) Weight maps corresponding to each iteration. (Row 3) AUC scores by separately using the weight map of each iteration. (Row 4) Images 1 and 3 are the anomaly maps obtained by the product rule and sum rule, respectively; images 2 and 4 images are the optimized weight maps corresponding to images 1 and 3; and image 5 is the final anomaly map.

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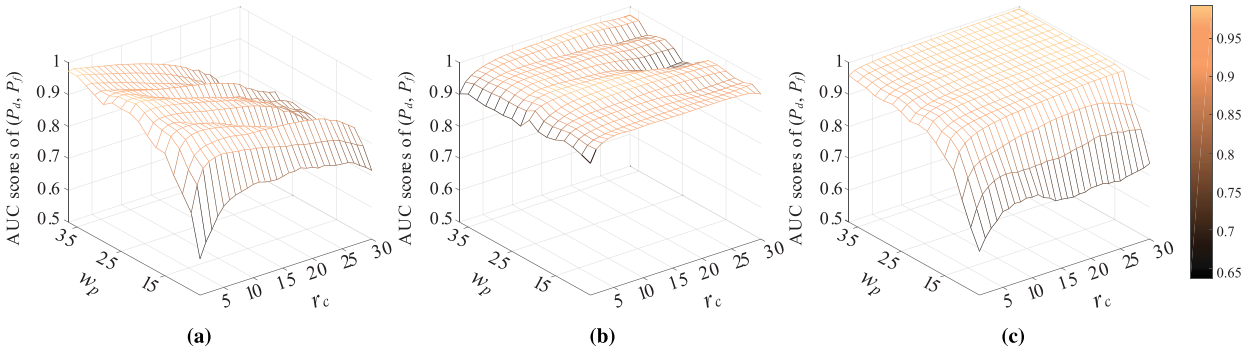


Fig. 12. Three types of variation trends of the AUC scores of (P_d, P_f) with w_p and r_c. Taking (a) ABU-airport (a), (b) ABU-beach (d), and (c) ABU-urban (d) as examples.

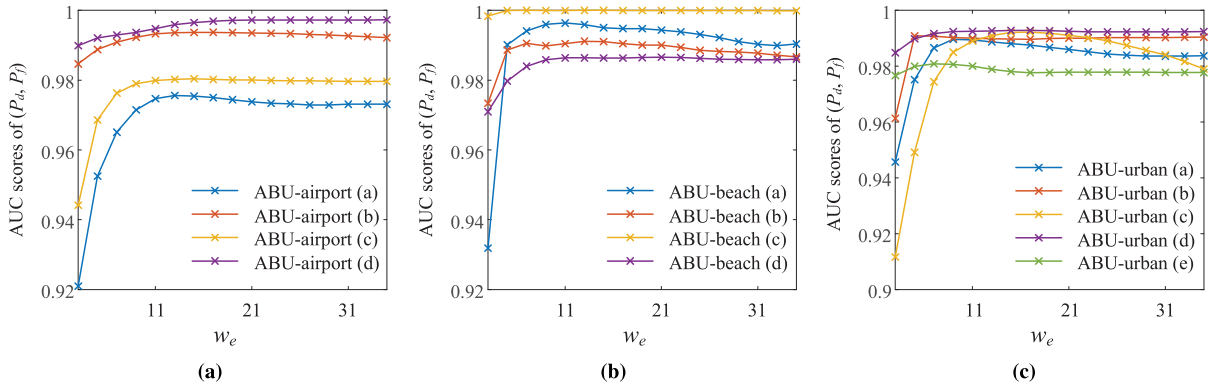


Fig. 13. Changes of AUC scores of (P_d, P_f) with w_e for each image of each scene: (a) ABU-airport scene, (b) ABU-beach scene, and (c) ABU-urban scene.

a significant CHE of these two detectors. In addition, taking these two factors into account, the EEMD is located before (or below, on other words) all five detectors. It is theoretically optimal that AUC score of (P_d, P_f) is 1 and CHE score is 0 at the same time.

In addition, we comprehensively compare the CHE score, AUC score of (P_d, P_f), AUC score of (P_f, τ), and ROC curves for the INPD, SED, JED, RED, and EEMD, averaged over all images of each scene, as shown in Figs. 9 and 10. Obviously, the EEMD always outperforms other detectors on each scene, in terms of each of the evaluation metrics. It fully proves the effectiveness of the integration strategy (over each individual entropy-based metric) and the generalization ability of the proposed EEMD.

3) Analysis for Weight Map: Here, taking the experiment on the ABU-airport-(c) as an example, we analyze the effectiveness and rationality of weight map $\mathcal{W}$. The first row of Fig. 11 shows the iteration process when optimizing $\mathcal{W}$. It can be observed that the function value (FV) converges after several iterations. The second row is the weight maps corresponding to each iteration. The third row of Fig. 11 shows the AUCs by separately using the weight map of each iteration. With the decrease of image standard deviation (FV), the improvement of AUC accuracy can be observed. When the function converges, the accuracy of AUC tends to be constant. This indicates the effectiveness of the standard deviation-based weight map optimization strategy. Furthermore, in the fourth row of Fig. 11, images 1 and 3 are the anomaly maps obtained by the product rule and sum rule, respectively, and images 2 and 4 are the optimized weight maps corresponding to images 1 and 3, respectively. According to the distribution of the two weight maps, it can be observed that the weight map tends to retain the nonanomaly region of image 1 and

TABLE II
OPTIMAL PARAMETERS SETTINGS FOR THE PROPOSED EEMD METHOD ACCORDING TO THE HIGHEST AUC SCORES OF (P_d, P_f)

Images\Parameters		w_p	r_c	w_e
ABU-airport	(a)	35	2	13
	(b)	33	8	17
	(c)	35	2	15
	(d)	15	17	35
ABU-beach	(a)	9	30	11
	(b)	31	10	13
	(c)	21	30	15
	(d)	19	11	21
ABU-urban	(a)	23	7	9
	(b)	29	30	5
	(c)	27	10	17
	(d)	35	30	17
	(e)	21	17	7

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TABLE III
AUC SCORES OF (P_d, P_f) ON THE ABU DATASET FOR DIFFERENT HYPERSPECTRAL ANOMALY DETECTION METHODS

Images\Methods		RX	LRX	LSMAD	CRD	DCR	MoG	EEMD	EEMD*
ABU-airport	(a)	0.8219	0.9666	0.8377	0.9512	**0.9668**	0.8398	0.9586	0.9767
	(b)	0.8404	0.9489	0.9195	0.9694	0.9460	0.8601	**0.9910**	0.9950
	(c)	0.9288	0.9410	0.9397	0.9580	0.9532	0.9125	**0.9759**	0.9850
	(d)	0.9526	0.9537	0.9873	0.9834	0.9346	0.9433	**0.9870**	0.9972
	Average	0.8859	0.9526	0.9211	0.9655	0.9501	0.8889	**0.9781**	0.9885
ABU-beach	(a)	0.9807	**0.9980**	0.9768	0.9891	0.9913	0.9908	0.9916	0.9968
	(b)	0.9106	0.9755	0.9054	0.9249	0.9590	0.9243	**0.9894**	0.9912
	(c)	**0.9999**	**0.9999**	0.9996	0.9997	**0.9999**	0.9998	**0.9999**	0.9999
	(d)	0.9538	0.9314	**0.9726**	0.8751	0.7919	0.8804	0.9650	0.9867
	Average	0.9613	0.9762	0.9636	0.9472	0.9355	0.9496	**0.9865**	0.9937
ABU-urban	(a)	0.9907	0.9807	0.9830	0.9851	0.9865	0.9846	**0.9891**	0.9895
	(b)	0.9946	0.8538	0.9840	0.9541	0.9662	**0.9848**	0.9575	0.9914
	(c)	0.9513	0.9459	0.9641	0.9579	0.9802	0.9305	**0.9829**	0.9924
	(d)	0.9887	0.8534	0.9814	0.9687	**0.9902**	0.9811	0.9825	0.9929
	(e)	0.9692	0.8811	0.9610	0.9529	0.9652	0.9525	**0.9773**	0.9807
	Average	0.9789	0.9030	0.9747	0.9637	0.9777	0.9667	**0.9779**	0.9894

the anomaly region of image 3. This can be explained as follows: 1) image 1 has considerable background suppression ability and image 3 has a higher anomaly score; 2) an expected anomaly map requires good background suppression and great anomaly emphasis; and 3) the combination strategy based on standard deviation can retain the background area of image 1 and the anomaly area of image 3, which shows the rationality of the strategy. In addition, the AUC scores of the final anomaly map are 0.9849, while those of the anomaly maps obtained by the product rule and sum rule are only 0.9776 and 9803, respectively.

D. Parameter Settings and Comparison Result

In this article, the proposed EEMD is compared with other widely used anomaly detection methods, including the GRX [13], LRX [13], LSMAD [44], CRD [45], DCR [67], and MoG [24]. GRX and LRX are typical methods for hyperspectral anomaly detection. The LSMAD is the low-rank and sparse matrix decomposition (LRaSMD)-based methods. The CRD is a representation-based detector. The DCR is an ensemble method that combines the inner window and outer window collaborative representations. The MoG is mixture of Gaussian-based method for accurately characterizing complex distribution.

1) Parameter Settings: In order to ensure the fairness and rationality of the experiment, fixed parameter settings are used for each competitor. For the LRX, CRD, and DCR, the sensitive parameters w_{in} and w_{out} are selected according to the best AUC scores averaged over the ABU dataset. Specifically, w_{in} and w_{out} are set to (11, 13), (5, 25), and (7, 15) for the LRX, CRD, and DCR, respectively. For the MoG and LSMAD, we use the default parameters suggested in [24] and [44], respectively. For the proposed EEMD, there are three parameters, i.e., w_p, r_c, and w_e. w_p and r_c are the block size and the cutoff distance controlling factor in the DPC process, respectively. w_e is the sliding window size in the entropy-based metrics. We set $w_p = 35$, $r_c = 8$, and $w_e = 13$ as the default parameter settings.

Although the fixed parameter settings for the proposed method are given, the effects of these parameters are analyzed. The values of w_p, r_c, and w_e range from 5 to 35, 1 to 30, and 3 to 35, respectively. When adjusting one parameter, the others are fixed. According to our observation, when w_p and r_c are analyzed together, there will be three different types of variation trends for different images, which can be observed in Fig. 12 that shows the changes of detection accuracies (the AUC scores of (P_d, P_f) is considered here) with w_p and r_c (take the ABU-airport (a), ABU-beach (d), and ABU-urban (d) images as examples). As it can be seen from Fig. 12(a) and (c), the detection accuracies increase with the increase of w_p. However, with the increase of r_c, the detection accuracy of Fig. 12(a) decreases rapidly, while that of Fig. 12(c) increases slowly. The case is different when looking at Fig. 12(b), where the detection accuracy is not sensitive to both w_p and r_c, and the optimal w_p and r_c appears in the middle area of the intervals of [5, 35] and [1, 30], respectively. In addition, the changes of detection accuracies with w_e for each image of each scene are shown in Fig. 13. It is apparent that with the increase of w_e, the detection accuracy is increased first and then stabilized for all images.

In addition, we report the optimal parameters settings used for the experiments in Table II. From the observations in Table II and the ground truth in Figs. 14–16, we can infer that there is a significant relationship between the optimal w_p and the size and quantity of anomaly targets. Anomalous targets with larger size or quantity usually lead to a larger value of the optimal w_p.

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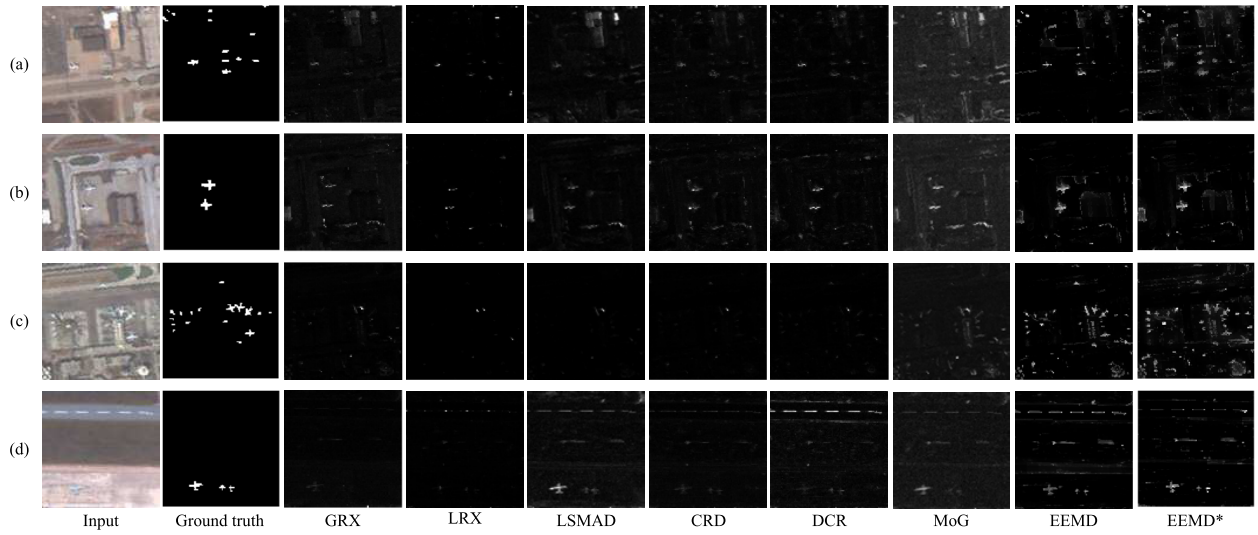


Fig. 14. Detection result maps by the GRX, LRX, LSMAD, CRD, DCR, MOG, and proposed EEMD methods for the ABU-airport scene. (a) ABU-airport-(a). (b) ABU-airport-(b). (c) ABU-airport-(c). (d) ABU-airport-(d).

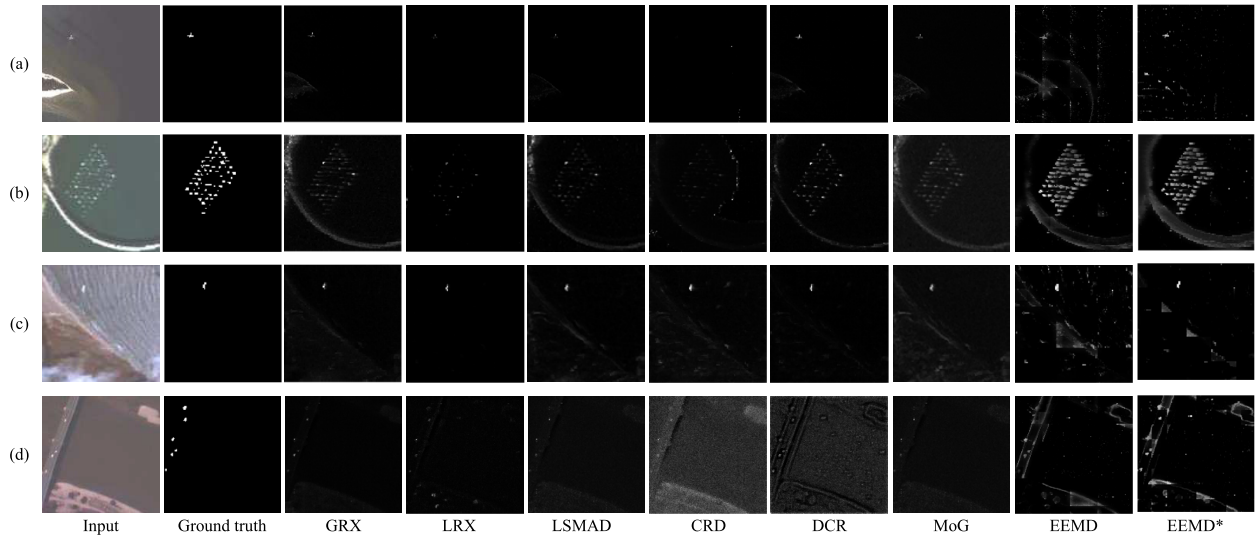


Fig. 15. Detection result maps by the GRX, LRX, LSMAD, CRD, DCR, MOG, and proposed EEMD methods for the ABU-beach scene. (a) ABU-beach-(a). (b) ABU-beach-(b). (c) ABU-beach-(c). (d) ABU-beach-(d).

2) Experiment on the ABU Dataset: A comparison between the proposed EEMD and other competitors is conducted on four aspects, i.e., the AUC scores of (P_d, P_f), the detection result maps, the ROC curves of (P_d, P_f), and the statistical separability analysis.

Tables III shows the AUC scores of (P_d, P_f) by different anomaly detection methods on each image (noting that, EEMD* refers to the EEMD using optimal parameters). On the whole, the average AUC scores of (P_d, P_f) by the EEMD over each of the three scenes are higher than other comparison methods, especially in airport and beach scenes, which are 0.0126 and 0.0229 higher than that of the second (obtained by the CRD and LSMAD, respectively). In addition, a significant improvement can be seen in comparison with other methods. For example, in the airport scene, the average AUCs of the RX and MoG are 0.8859 and 0.8889, respectively, while that of the proposed EEMD is 0.9781. Moreover, the EEMD using optimal parameters can outperform all other methods considered. Therefore, we are able to conclude from Table III that, benefitted from the ensemble entropy metric strategy we proposed, the EEMD can achieve a quite competitive detection performance.

The result detection maps obtained by different methods are shown in Figs. 14–16. As can be seen visually, the EEMD can better recover anomalies of different sizes and shapes by suppressing background and highlighting anomalies. For instance, on whether the ABU-airport scene or the ABU-beach scene, the anomalies can be correctly detected by the EEMD method. However, the RX and LRX methods generally do not show detection capabilities on most of the considered images, such as in Fig. 14(c) and (b), and we can hardly see the anomalies visually. The MoG method performs terrible background suppression and tends to emphasize background, leading to high false alarm rates, which can be observed from

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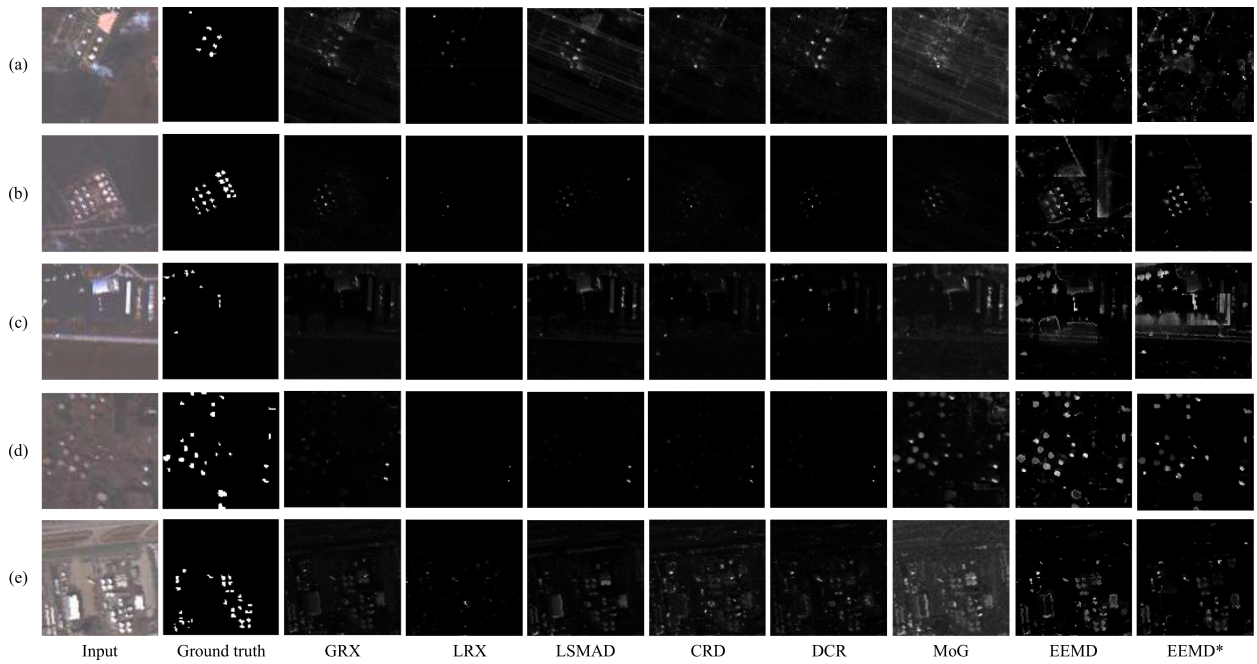


Fig. 16. Detection result maps by the GRX, LRX, LSMAD, CRD, DCR, MOG, and proposed EEMD methods for the ABU-urban scene. (a) ABU-urban-(a). (b) ABU-urban-(b). (c) ABU-urban-(c). (d) ABU-urban-(d). (e) ABU-urban-(e).

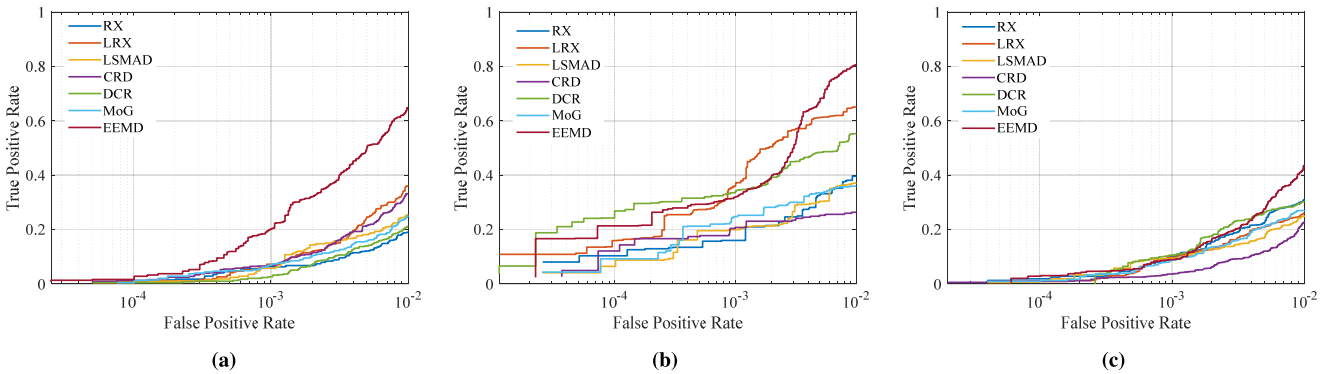


Fig. 17. ROC curves of (P_d, P_f) for different methods, averaged over each image of each scene: (a) ABU-airport scene, (b) ABU-beach scene, and (c) ABU-urban scene.

the airport and urban scene. Although the CRD and DCR have detection ability to some extent, they cannot effectively preserve the shape of abnormal targets and generates some miss detection. By contrast, the proposed EEMD can effectively detect anomalies of different shapes and sizes. For example, the anomalies are detected and structural information of the anomaly target (aircrafts) in Fig. 14 is very clearly preserved by our EEMD.

Fig. 17 shows the ROC curves of (P_d, P_f) obtained by different anomaly detection methods, averaged over each scene of the ABU dataset. In Fig. 17, we show all ROC curves for $P_f < 10^{-2}$ to assess the performance of the proposed method. From Fig. 17, we can see that the EEMD performs much better than comparative methods in most cases of false positive rate on the airport and urban scenes. In the beach scene, although the EEMD performs worse than the DCR when $P_f < 10^{-3}$, its P_d achieves 0.8 when $P_f = 10^{-2}$.

To further evaluate the separability between the anomaly and background, the separability maps are employed for comparison with the other methods. Excluding the biggest 20% and the smallest 20% values of normalized detection results of anomaly pixels and background pixels, boxes are drawn to enclose the middle 60% of the main pixels. In Fig. 18, the red boxes and green boxes represent the distribution of the detection values of anomaly and background pixels, respectively. The line inside the boxes is the median of the pixels' values. The position of the boxes reflects the separability of the anomalies and background. In other words, the larger the distance between the median line of the two boxes, the better the method is. As shown in Fig. 18, the distance between the median of the red and green boxes for the proposed EEMD is larger than any of other detectors, illustrating a superior discriminative of the EEMD.

Table IV reports the computational time of the comparison detectors on the ABU dataset. The experiments were

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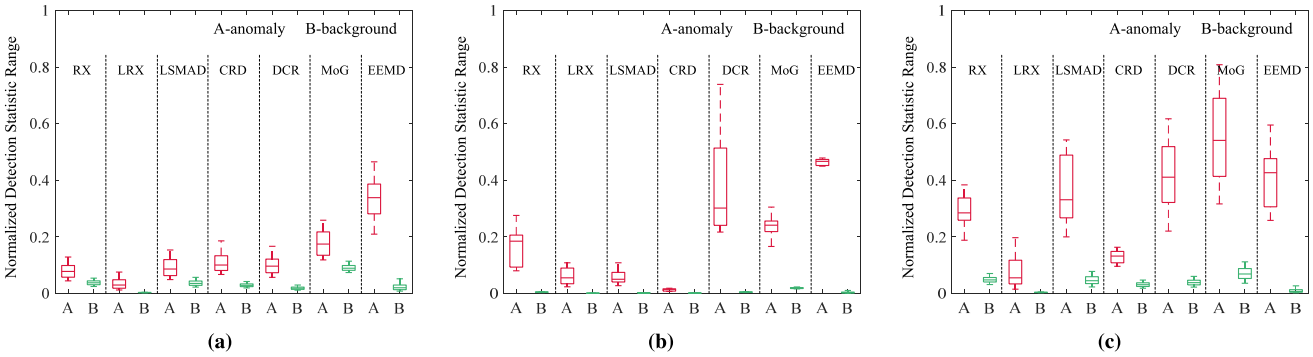


Fig. 18. Separability maps of different methods on (a) ABU-airport-(a), (b) ABU-beach-(a), and (c) ABU-urban-(a).

TABLE IV
EXECUTION TIME (IN SECONDS) ON THE ABU DATASETS BY DIFFERENT COMPARED METHODS

	A-(a)	A-(b)	A-(c)	A-(d)	B-(a)	B-(b)	B-(c)	B-(d)	U-(a)	U-(b)	U-(c)	U-(d)	U-(e)
GRX	0.1	0.1	0.1	0.2	0.6	0.1	0.1	0.1	0.3	0.1	0.1	0.1	0.1
LRX	47.6	47.8	46.9	42.3	171.7	40.6	39.3	28.8	45.7	48.4	40.7	47.4	48.0
LSMAD	11.9	11.2	12.0	10.6	25.2	11.0	10.7	13.8	11.7	11.7	11.0	11.6	11.5
CRD	578.6	489.8	795.7	861.7	1694.1	522.6	496.4	1704.8	521.3	526.4	512.6	522.9	530.7
DCR	91.6	83.5	152.8	133.6	181.9	81.0	83.2	201.9	101.3	86.3	83.3	85.5	86.7
MoG	13.5	7.9	9.5	11.9	39.1	21.0	13.7	13.6	20.0	24.0	14.1	16.5	15.0
EEMD	10.6	11.6	11.3	7.6	21.1	10.5	7.8	20.6	11.7	7.9	11.0	11.7	9.5

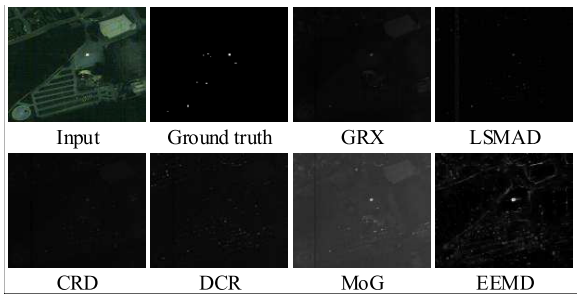


Fig. 19. Detection result maps by the GRX, LRX, LSMAD, CRD, DCR, MOG, and proposed EEMD methods on the Viareggio dataset.

TABLE V
AUC SCORES OF (P_d, P_f) AND EXECUTION TIME ON THE VIAREGGIO DATASET FOR DIFFERENT HYPERSPECTRAL ANOMALY DETECTION METHODS

	RX	LRX	CRD	DCR	MoG	EEMD
AUC	0.8593	0.9568	0.9609	0.9532	0.9280	0.9698
time/s	4.9	5649.0	17196.9	2882.1	1176.7	155.1

implemented in MATLAB 2019a on a computer with a 3.5-GHz CPU and a 16-GB RAM. Obviously, from Table IV, except slower than GRX, the EEMD is much faster than LRX, CRD, and DCR and is slightly faster than MoG.

3) Experiment on the Viareggio Dataset: The proposed method is further evaluated on the Viareggio dataset. Since the size and number of bands of the Viareggio dataset are much larger than those of the ABU dataset, the parameters used for the Viareggio dataset should be different. For the LRX, CRD, and DCR, w_{in} and w_{out} are set to (7, 25), (7, 27), and (5, 21), respectively. For the MoG, the default parameters suggested in [24] are used. For the EEMD, we set $w_p = 39$, $r_c = 1$, and $w_e = 25$ here. Fig. 19 shows the detection maps obtained by different methods. As the size of the image increases, the proportion of anomaly targets decreases, which increases the difficulty of detection. All the comparison methods, including EEMD, can only detect the largest abnormal target effectively. In addition, compared with other methods, EEMD can highlight the anomaly most. Table V shows the AUC scores and execution time for different detectors. The EEMD performs best compared to other methods, whose AUC is 0.9698. According to the execution time, the EEMD is much faster than other methods accepted RX.

V. CONCLUSION

This article proposed a new EEMD for hyperspectral anomaly detection, which is based on the entropy analysis of the system in information theory. The main idea of the proposed method is transforming the system entropy into three metrics evaluating the anomaly score of pixels, by considering the SE, JE, and RE and correspondingly establishing the systems. Our ensemble strategy for the three individual entropy-based metrics can not only improve the detection rate but also reduce the false alarm rate. Our experimental results, conducted on the ABU dataset and the Viareggio dataset, demonstrate the superiority of the proposed method with regard to state-of-the-art competitors. As with any new approach, there are

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still some unresolved issues that may present challenges over time. Specifically, the proposed method still produces some false alarms in the detection result. These aspects deserve further research and improvements in future work. We are also planning on developing an efficient implementation of the proposed approach by resorting to high-performance computing architectures such as graphics processing units (GPUs).

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